

On the estimation of specific surface per unit pore volume from induced polarization: A robust empirical relation fits multiple data sets

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ABSTRACT

We analyze the relationship between induced polarization (IP) parameters and the specific surface area normalized to the pore volume (S_{por}) for an extensive sample database. We find that a single linear imaginary conductivity- S_{por} relation holds across a range of single-frequency IP data sets composed of sandstones and unconsolidated sediments that lack an appreciable metallic mineral content. We also apply a recent approach defined as Debye decomposition (DD) to determine normalized chargeability (m_n), a global estimate of polarization magnitude from available spectral IP (SIP) data sets. A strong linear relationship between m_n and S_{por} is also found across multiple data sets. However, SIP model parameters determined for samples containing metallic minerals are approximately two orders of magnitude greater than for the model parameters estimated for the nonmetallic sample database. We propose a concept of “polarizability of the mineral-fluid interface per unit S_{por} ” to explain this difference, which is supported by the observed dependence of IP parameters on fluid conductivity between sample types. We suggest that this linear IP- S_{por} relation can be considered the IP equivalent of the classical Archie empirical relation. Whereas the Archie relation describes a power-law relation between electrical conductivity due to electrolytic conduction through the available interconnected pore volume, the IP- S_{por} relation is an equivalent relation between mineral-fluid interfacial polarization and available pore surface area.

INTRODUCTION

Induced polarization (IP) is a mature geophysical prospecting technique with applications in mining and environmental exploration. Although the theoretical principles of the technique with re-

spect to mineral exploration have long been established (e.g., the seminal work of Wait 1982 or the comprehensive review by Luo and Zhang 1998), the significance of the technique in environmental research is still emerging (Börner et al., 1993; Weller and Börner, 1996; Slater and Lesmes, 2002). Although the dependence of the IP effect on the polarization of metallic mineral surfaces has long been recognized (e.g., Pelton et al., 1978; Wong, 1979), it is the more recent hydrogeophysical research on the IP method that has repeatedly shown that IP data can be correlated with physical properties of the pore space in nonmetallic soils and rocks. Börner and Schön (1991) first report a linear dependence of the IP response on the specific surface area normalized to the total volume (S_{tot}) for a range of sandstone samples. Börner et al. (1996) show that this relation can be improved if the specific surface area is normalized to the pore volume (S_{por}), a parameter frequently used to characterize the transport properties of porous media (e.g., Riepe et al., 1979) and often closely associated with the cation exchange capacity (controlling reactive transport and soil chemistry). The improvement in the relations is a result of the wide range of porosity encountered in soils and rocks. Subsequent studies conducted on unconsolidated sediments have reported strong power-law relations between IP parameters and S_{por} with exponents between 0.8 and 1.2 (e.g., Slater and Glaser, 2003; Slater et al., 2006; Slater, 2007). However, recent studies conducted on sandstone samples with a limited range of S_{por} variation have reported weaker relationships between S_{por} and IP measurements (Binley et al., 2005; Scott and Barker, 2005).

In this paper, we re-examine the IP- S_{por} relationship using a database compiled from the published literature and new data sets (sandstone and unconsolidated sediments). One complication in evaluating the IP- S_{por} relationship between different data sets results from the fact that reported IP parameters have varied between the published studies. Some authors have reported single-frequency IP parameters (e.g., the imaginary part of the complex conductivity σ'' at a single frequency) whereas others have reported parameters describing the polarization magnitude retrieved by modeling the frequency dependence of spectral data sets (e.g., the global polarization magni-

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tude from a Cole-Cole or generalized Cole-Cole model). To assess the nature of the $IP-S_{por}$ relation across a broad range of data sets, we apply a recent approach defined as Debye decomposition (DD; Appendix A; Nordsiek and Weller, 2008), to determine a new parameter defining the global polarization magnitude from available spectral IP (SIP) data sets. The DD can be applied to any spectral data set whether the spectra exhibit constant phase, Cole-Cole, or some other behavior.

We show that a single linear $IP-S_{por}$ relation holds across a range of available data sets composed of sandstones and unconsolidated sediments lacking appreciable metallic mineral content. Consistent with previous work (e.g., Vinegar and Waxman, 1984; Lesmes and Frye, 2001; Slater and Glaser, 2003), we find only a weak dependence of IP magnitude on fluid conductivity. We also present evidence that a linear $IP-S_{por}$ relationship also exists in the case of electronic conduction associated with metallic particles. However, the gradient of the $IP-S_{por}$ relationship is steeper, which we attribute to greater reactivity of the metal-fluid interface relative to the nonmetal-fluid interface.

IP parameters and data sets

Induced polarization measures an electrochemical polarization associated with the electrical double layer (EDL) forming at the min-

eral-fluid interface of soils and rocks. Various conceptual and theoretical models exist to describe this phenomenon. Some models consider the polarization of the EDL itself surrounding a mineral grain (e.g., Lesmes and Morgan, 2001) whereas others consider the polarization of pore throats due to differences in ionic mobility between wide and narrow pores (Marshall and Madden, 1959; Titov et al., 2002). Considering EDL polarization, it is conceptually reasonable to assume that measures of the magnitude of the polarization will scale with the amount of available interfacial surface subjected to an applied electric field. Although this is not so conceptually obvious for the polarization of pore-throat model, the numerous studies investigated here lend support to the concept of a broad $IP-S_{por}$ relation.

All laboratory SIP and IP measurements reported in Tables 1 and 2 of this paper were obtained by measuring the impedance magnitude and phase shift of the voltage waveform recorded across the sample relative to the current waveform recorded on a reference resistor, the source typically being a sine signal (e.g., Slater and Lesmes, 2002). All measurements reported here were recorded using a four-electrode device whereby separate electrode pairs were used to inject the current into the sample and record the potential waveform across the sample. Measured magnitude and phase were converted into a measured complex conductivity (σ^*) or complex resistivity (ρ^*) using

Table 1. Compilation of data sets to display the relation between imaginary part of conductivity (σ'') measured at a frequency of 1 Hz and the pore-space-related internal surface S_{por} . Power-law (equation 1) and linear (equation 2) fitting parameters are shown.

Sample set	Origin	# of samples	Solution	Brine conductivity (mS/m)	Equation 1				Equation 2		References
					a_p	b_p	R^2	rms	c_p	rms	
Sandstone A	Germany, France	19	NaCl	100	0.0152	0.922	0.964	0.154	0.0137	0.168	Börner and Schön, 1991
Sandstone B	U. K.	13	NaCl	90	0.0014	1.870	0.844	0.260	0.0126	0.382	Scott, 2003
Sandstone C	U. K.	14	Synthetic ground-water	100	0.1188	0.375	0.362	0.095	0.0143	0.153	Binley et al., 2005
	U.S.A.	1	NaCl	100							Lesmes and Frye, 2001
Sandstone D					0.0191	0.719	0.771	0.150	0.00888	0.184	
	Germany	3	NaCl	88–114							Kruschwitz, 2008
	Germany, U. K.	6	NaCl	90							Breede, 2006
	Germany	3	NaCl	90							Schröder, 2008
	Germany, U.S.A.	4	NaCl	100							new data
Sandstone E	Bahariya Fm., Egypt	24	NaCl	90–100	0.0247	0.062	0.009	0.128	0.00637	0.227	new data
All Sandstones		87			0.0103	0.992	0.822	0.271	0.0101	0.271	
Sand	Olathe, KA U.S.A.	12	NaCl	60	0.00496	1.263	0.556	0.209	0.00715	0.214	Slater and Glaser, 2003
Sand-clay mixture		12	NaCl	100							Slater et al., 2006
					0.0113	1.023	0.914	0.132	0.0115	0.133	
		3	NaCl	100							new data
All sand samples		27			0.0101	0.916	0.756	0.202	0.00931	0.202	
All samples		114			0.0100	0.996	0.840	0.257	0.00993	0.257	

the geometric factor defining the measurement geometry and the resulting current flow path in the test device. This corresponds to the field-scale frequency domain IP or SIP technique.

Interpretation of IP data sets across published studies is complicated by the fact that different IP parameters (depending on whether the measurements are single frequency or multifrequency and whether the data are frequency domain or time domain) have been used to report results. In this paper, we report results for two IP parameters. The first is the imaginary conductivity (σ''), which is a frequency-dependent measure of the polarization magnitude calculated from a measured magnitude and phase term recorded in the frequency domain. In compiling this database, we chose to focus on frequencies close to 1 Hz because our interest is in the interpretation of IP data sets (particularly field measurements) that are recorded at low frequencies. Furthermore, additional polarization mechanisms (e.g., Maxwell-Wagner interfacial polarization) become important at higher frequencies. These polarization phenomena are usually not

recorded in field IP data sets, in which usable frequencies are often restricted to <10 Hz due to excessive electromagnetic coupling at higher frequencies. One benefit of considering σ'' as our first IP parameter is that more published data sets are available when compared to SIP data sets that capture the frequency dependence of the response. Furthermore, the presence of a robust IP- S_{por} relation for a single-frequency measurement is attractive because it could be applied to field IP data sets that are more routinely collected than field SIP data sets. The database used here is summarized in Table 1. It consists of more than 100 samples originating from nine published studies (seven are sandstone data sets; two are unconsolidated sediment data sets) and a previously unpublished data set (sandstone E) from the Bahariya Formation (Egypt). Halisch et al. (2009) give a general description of the origin of the Bahariya samples. Only the isotropic samples have been included in this study. The data set sandstone D is a compilation of samples of different origins to reach a sufficient number of points to calculate the correlation line. To mini-

Table 2. Compilation of data sets to display the relation between normalized chargeability m_n derived from Debye decomposition and the pore-space-related internal surface S_{por} . Power-law (equation 4) and linear (equation 5) fitting parameters are shown.

Sample set	Origin	# of samples	Solution	Brine conductivity (mS/m)	Equation 4				Equation 5		References
					a_s	b_s	R^2	rms	c_s	rms	
Sandstone B	U. K.	13	NaCl	90	0.02373	1.576	0.836	0.225	0.1018	0.292	Scott, 2003
Sandstone C	U. K.	14	Synthetic ground-water	100	1.078	0.247	0.280	0.076	0.0844	0.163	Binley et al., 2005
	U.S.A.	1	NaCl	100							Lesmes and Frye, 2001
Sandstone D	Germany	3	NaCl	88–114	0.181	0.641	0.768	0.134	0.0679	0.192	Kruschwitz, 2008
	Germany, U. K.	6	NaCl	90							Breede, 2006
	Germany	3	NaCl	90							Schröder, 2008
	Germany, U.S.A.	4	NaCl	100							new data
Sandstone E	Bahariya Fm., Egypt	24	NaCl	90–100	0.228	0.023	0.001	0.121	0.0557	0.230	new data
All Sst. samples		68			0.0776	0.982	0.757	0.233	0.0716	0.244	
Sand-clay mixture		12	NaCl	100							Slater et al., 2006
					0.152	1.178	0.852	0.207	0.168	0.220	
		3	NaCl	100							new data
Sand + iron		9	NaNO ₃	100	1.829	1.175	0.946	0.168	2.334	0.199	Slater et al., 2006
Sand + magnetite		9	NaNO ₃	100	3.345	1.041	0.969	0.105	3.515	0.108	Slater et al., 2006
Sand + iron or magnetite		18			2.508	1.102	0.933	0.173	2.864	0.183	

mize any effects of variable pore-fluid chemistry between data sets, we report measurements for a narrow range of fluid conductivity (σ_w) (0.060–0.114 S/m).

The single-frequency analysis will neglect the effects of the shape of the frequency dependence of σ'' on any resulting IP- S_{por} relation. Changes in σ'' at a specific frequency can result from changes in the shape of the frequency dependence. Common physicochemical models used to describe the shape of the dispersion include the constant phase model when spectra are close to flat (e.g., Börner et al., 1996) and the Cole-Cole model (or its variants) when the shape is described by a broad range of relaxation times centered on a peak in the phase spectra. The choice of IP parameter used to interpret IP- S_{por} relations has, hence, varied between studies depending on the shape of the spectra and the model subsequently adopted to describe it. However, phase spectra reported in previous studies show a broad range of shapes. Figure 1a shows phase spectra of sandstone samples from different locations within Europe. Sample BR5 shows a typical Cole-Cole behavior with an accentuated phase peak. In contrast, the phase spectra of samples OK4 and BK are flat and do not show any phase peak. Figure 1b demonstrates the variability of phase spectra within a single geologic formation in which not all the spectra can be fit to Cole-Cole type models. Sample B51H shows a particularly in-

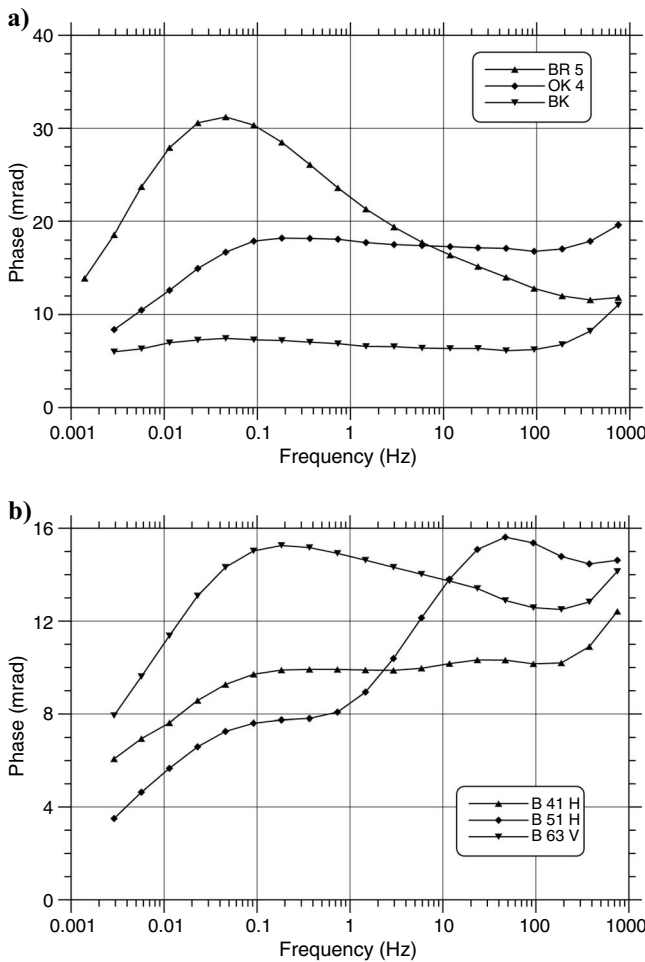


Figure 1. Phase spectra of sandstone samples. (a) Samples originating from different formations. (b) Samples from the Bahariya Formation, Western Desert, Egypt.

teresting shape that is somewhat similar to the GEMTIP model based on a recent generalized effective medium theory for IP modeling (Zhdanov, 2008).

To facilitate the assessment of the IP- S_{por} relationship from the database of available SIP data sets, we adopt a recent modeling approach defined here as DD, which can be applied to any SIP data set irrespective of the shape of the spectra (Nordsiek and Weller, 2008). The DD fits spectra to a broad range of relaxation times that are computed for a distribution of Debye relaxations with a range of discrete time constants spanning the measured frequency range. The polarization magnitudes computed for each individual Debye relaxation are integrated across the frequency range to calculate a global polarization magnitude term, defined here as normalized chargeability (m_n), which accounts for the variable shapes of the spectra. The normalized chargeability is the product of total chargeability and the low-frequency conductivity that are two out of four integrating parameters resulting from the DD fitting. This normalization of the total chargeability by multiplying it with the low-frequency conductivity is equivalent to the calculation of the metal factor from measurements of the percentage frequency effect used in mineral exploration (Lesmes and Frye, 2001). A detailed discussion of the DD approach is beyond the scope of this study but can be found in Nordsiek and Weller (2008). A summary of the most relevant formulas is given in Appendix A.

The SIP database is reported in Table 2. It consists of 68 sandstone samples from six published studies and a set of 24 new samples from the Bahariya Formation (Egypt). Fifteen sand-clay mixtures are also reported. We also include a set of 18 previously published sand-metal mixtures to explore the effects of variable surface reactivity on the IP- S_{por} relation. Note that the sand-metal mixtures are not included in the single-frequency data set (Table 1) because they exhibit much stronger frequency dependence (a result of the metallic polarization) than the nonmetallic samples; therefore, we consider it inappropriate to analyze the IP- S_{por} relation at a single frequency for these samples (as discussed in Slater et al., 2006). We again use as narrow a range of σ_w as possible (0.088–0.114 S/m) in our analysis.

Results

The σ'' - S_{por} relation

Table 1 summarizes the fitting parameters a_p and b_p for the power-law relation

$$\sigma'' = a_p (S_{\text{por}})^{b_p}, \quad (1)$$

and parameter c_p for the linear relation

$$\sigma'' = c_p (S_{\text{por}}), \quad (2)$$

when applied to individual data sets, all sandstone data sets, all unconsolidated sand data sets, and the entire data set (subscript p signifies a single-frequency IP data set). The fitting parameters for any combined relationship reported in Table 1 are independently determined as a single fit to all of the data points making up that combined relationship (i.e., the fitting parameters for the combined relationships do not represent a mean estimate calculated from the fitting parameters of individual data sets). The reported parameters are for S_{por} in μm^{-1} and σ'' in mS/m. All the given fitting parameters refer to the log transformed values of S_{por} and σ'' . The coefficient of determination R^2 is a widely acknowledged parameter to assess the quality of the fitting of relations by linear regression. Here, R^2 is defined as a

measure of the proportion of variability in the data accounted for by the given model and quantifies how well values of the IP parameters for new S_{por} data are likely to be predicted from the given model (in the specific case of the linear regression applied here, R^2 is also the square of the sample correlation coefficient). In the case of the linear equation 2, only one free parameter (the slope c_p) must be determined. Because the fitting procedure in a double logarithmic scale is not a linear regression, R^2 cannot be used to assess the fitting quality. The scatter around the fitting line can instead be well assessed by the rms error applied to the residual between the fit value and the actual data value of the imaginary part of conductivity at logarithmic scale

$$\text{rms} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\log_{10}(\sigma''_{i,\text{fit}}) - \log_{10}(\sigma''_{i,\text{data}}))^2}, \quad (3)$$

with n being the number of samples in the data set. If the rms error reaches unity, the average deviation from the fitting line is one logarithmic decade or a factor of 10.

Although there is significant variation in the fitting parameters between data sets, we find that the single power law with an exponent of 0.996 ($R^2 = 0.840$, rms = 0.257) and linear (rms = 0.257) relations can be fit to the entire data set. Figure 2 shows a plot of 114 samples with the single linear relation $\sigma'' = 0.01S_{\text{por}}$ obtained from this data set superimposed.

The m_n - S_{por} relation

Table 2 summarizes the fitting parameters a_s and b_s for the power-law relation

$$m_n = a_s(S_{\text{por}})^{b_s}, \quad (4)$$

and parameter c_s for the linear relation

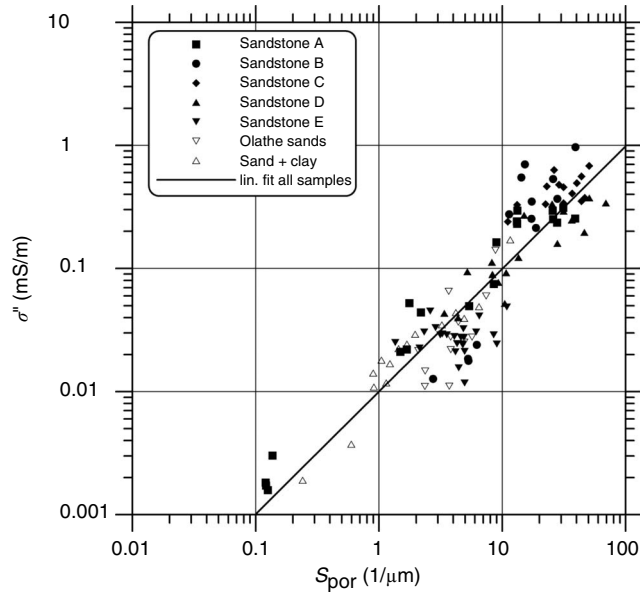


Figure 2. Relation between imaginary part of conductivity (σ'') measured at a frequency of 1 Hz and the pore-space-related internal surface S_{por} for 114 samples originating from nine data sets.

$$m_n = c_s(S_{\text{por}}), \quad (5)$$

when applied to individual data sets, all sandstone data sets, and all sand-metal data sets (subscript s refers to an SIP data set). The fitting parameters for any combined relationship reported in Table 2 are independently determined as a single fit to all of the data points making up that combined relationship (i.e., the fitting parameters for the combined relationships do not represent a mean estimate calculated from the fitting parameters of individual data sets). The reported parameters are for S_{por} in μm^{-1} and m_n in mS/m. The coefficients of determination R^2 and the rms error are used according to the previous explanation. Although there is significant variability between the fitting parameters when applied to individual data sets, we find that the single power law with an exponent of 0.982 ($R^2 = 0.757$, rms = 0.233) and linear (rms = 0.244) relations can be fit to all 68 sandstone samples. Although the combined metal samples also show a strong power law ($R^2 = 0.933$, rms = 0.173) and linear (rms = 0.183) relations, parameter a_s (equation 4) and c_s (equation 5) are approximately two orders of magnitude larger than those for the nonmetallic samples (Table 2). Figure 3 shows the SIP data sets with the linear fits for all sandstone samples ($m_n = 0.072S_{\text{por}}$), the unconsolidated sand-clay mixtures ($m_n = 0.17S_{\text{por}}$), and all sand-metal mixtures ($m_n = 2.86S_{\text{por}}$). We chose to plot separate relations for the sandstone and sand-clay mixtures to graphically show how c_s (the gradient of the linear relation) is largest for the metallic samples and c_s for the sand-clay mixtures is more than two times larger than for the sandstone samples. The significance of this variation in c_s between these data sets is discussed below.

DISCUSSION

We have shown that a single σ'' - S_{por} relation (Figure 2) can be fit to a database of sandstone and unconsolidated sediments that incorporates 114 samples from nine independent studies (Table 1). The fact that unconsolidated soils and consolidated sandstones follow a sin-

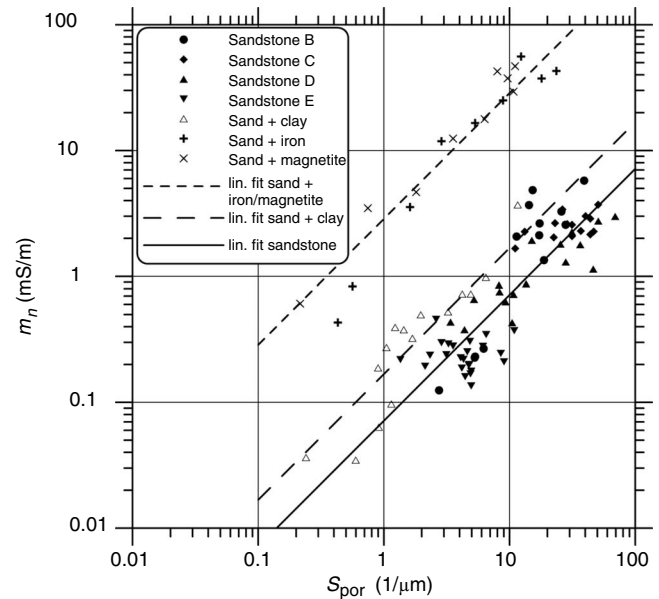


Figure 3. Relation between normalized chargeability (m_n) derived from DD and the pore-space-related internal surface S_{por} for different data sets.

gle relationship suggests that single-frequency IP measurements will primarily be a function of S_{por} despite the variation in other lithological factors within this database (e.g., degree of compaction, distribution of clays). This sensitivity to the interfacial surface makes IP a unique geophysical imaging technique with obvious potential for investigating lithologic variation (e.g., Kemna et al., 2004), hydraulic properties (e.g., Weller and Börner, 1996; Hördt et al., 2007), and properties controlling reactive transport. Furthermore, we did not find a substantial improvement in the IP- S_{por} relation for m_n estimated from SIP data sets compared to σ'' obtained at 1 Hz. This suggests that single-frequency IP measurements might be sufficient for empirical estimation of S_{por} and that (the much more time-demanding) SIP data acquisition will not necessarily improve the S_{por} estimation.

Although Binley et al. (2005) report no significant relation between σ'' and S_{por} , their samples do follow the multi-data set σ'' - S_{por} relation demonstrated here (Figure 2). Whereas our multistudy database considers three orders of magnitude of S_{por} variation, the S_{por} variation reported in Binley et al. (2005) is restricted to $11\text{--}50\text{ }\mu\text{m}^{-1}$ so that the σ'' - S_{por} relation is not strong based on their data set alone. The single IP- S_{por} relationship we report here thus appears applicable for determining the order of magnitude variability in S_{por} . However, we have not considered the possible correlation between S_{por} and other SIP model parameters such as those that quantify the relaxation time of the dispersion. For example, Binley et al. (2005) did observe a strong relationship between the time constant (τ) of a fitted Cole-Cole model and S_{por} for their database of 17 sandstone samples within a narrow range of S_{por} . They also found a strong correlation between τ and pore-throat diameter determined from mercury intrusion and a strong correlation between τ and hydraulic conductivity. We note that their spectra all show a Cole-Cole behavior with a characteristic peak in the phase spectrum from which τ was well constrained. Kruschwitz et al. (2010) explores the relationship between τ and pore diameter for multiple independent data sets

with results that show promise for improving the prediction of hydraulic properties from SIP spectra.

Next, we further consider the significance of our observation that the parameter c_s of the m_n - S_{por} relation is approximately two orders of magnitude larger for metallic samples than for nonmetallic samples (Figure 3). Here we invoke a concept of “polarizability of the mineral-fluid interface per unit S_{por} ” as a possible explanation of this difference. We argue that the presence of metallic minerals results in higher polarizability per unit S_{por} due to the large differences in the electrochemistry between the polarization of metallic interfaces (electronic conductors) versus nonmetallic interfaces. The nearly twice greater polarizability per unit S_{por} observed for clay-sand mixtures relative to the sandstone samples could be related to a higher polarizability of the kaolin powder used for the clay-sand mixtures when compared to the different clay mineral aggregates present in the sandstone samples.

To explore this concept further, we plot σ'' as a function of σ_w for representative samples from the sandstone, unconsolidated sand-clay mixtures, and metallic data sets (Figure 4). Excluding the metallic data sets, the response is characterized by an increase in σ'' with σ_w until a plateau is reached, beyond which σ'' is constant or decreases slightly with increasing σ_w . This behavior has been reported previously and is attributed to competing effects of increases in surface-charge density dominating at low σ_w and decreasing surface ionic mobility dominating at higher σ_w (e.g., Lesmes and Frye, 2001; Slater and Glaser, 2003). Figure 4 shows that the plateau of this curve appears to shift to higher σ_w progressing from the sandstone (lowest c_s parameter) to the metallic samples (highest c_s parameter) where the plateau is not reached over the range of measured σ_w . Furthermore, the gradient of the increase in σ'' with σ_w appears to increase, progressing from the sandstone samples to the metallic samples. We suggest that the stronger dependence of σ'' on σ_w for the metallic samples — and to a lesser extent sand samples containing clay (kaolinite) minerals — relative to the sandstone samples is a manifestation of this greater polarizability of the mineral-fluid interface per unit S_{por} for these samples. Under this conceptual model, the metallic and clay interfaces are able to support more charge per unit S_{por} before the polarization of the interface is saturated by excess charge accumulation on surfaces. Although further work is clearly needed to demonstrate this concept, it is consistent with the observation that metallic samples exhibit the same linear m_n - S_{por} behavior, differing only in the value of the parameters describing the slope of the S_{por} dependence.

Our findings, along with the concept of polarizability of the mineral-fluid interface per unit S_{por} , might shed some light on the relative importance of EDL polarization versus pore-throat-polarization mechanisms in porous rocks (Blaschek and Hördt, 2009). The fact that a simple linear or power-law relation between measures of polarization magnitude and S_{por} can be fit across multiple data sets is conceptually most consistent with a surface EDL polarization model and not the pore-throat-polarization model. It is quite possible that this relationship would break down in samples where pore-throat polarization dominates because this polarization is more likely controlled by differences in ionic mobility between pores of different sizes. Consequently, rather than suggesting that the relationships between polarization magnitude and S_{por} determined here hold for all soils/rocks, we consider these relationships valid for soils/rocks in which EDL polarization is likely to dominate over pore-throat polarization. Similarly, the concept of polarizability of the mineral surface per unit S_{por} is expected to hold when EDL polarization is domi-

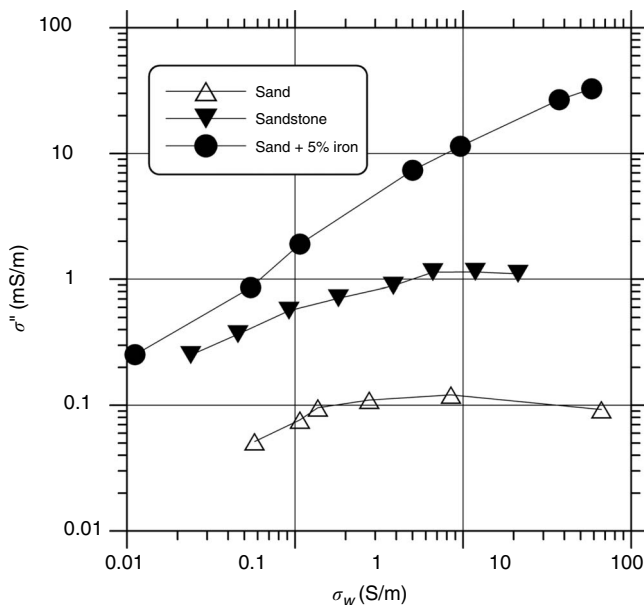


Figure 4. Relation between the conductivity of the saturating sodium chloride solution and the imaginary part of conductivity (σ'') for samples of sand (N2 37-35, Slater and Glaser, 2003), sandstone (sample GR; Breede, 2006), and a sand-iron mixture (Ottawa sand + 5% iron; Slater et al., 2006).

nant. This is most likely the case for unconsolidated soils and loosely consolidated rocks in which abrupt changes in pore diameter are less likely. It is such soils/rocks that make up the experimental database that was used here. We recognize that the validity of this concept of polarizability of the mineral-fluid interface per unit S_{por} needs further investigation, especially outside of the range of S_{por} and σ_w explored here.

CONCLUSION

Using the most comprehensive database currently available to us, we have demonstrated the existence of a linear IP- S_{por} relation for sandstone and unconsolidated soils. The relation describes IP measurements around 1 Hz for a database composed of more than 100 samples. Using a DD approach applicable to phase spectra of any shape, we also report the IP- S_{por} relation observed for SIP data sets, allowing us to incorporate an analysis of the effects of metallic particles on the IP- S_{por} relation. We find that the same linear dependence of polarization magnitude on S_{por} exists for metallic particles but that parameters describing the gradient of the IP- S_{por} relation are larger. We suggest that the gradients are related to the polarizability of the mineral-fluid interface per unit S_{por} , which is controlled by the electrochemistry of the polarization. From a physical viewpoint, a linear relation between polarizing internal surface and IP effect can be expected. Scatter in the data points around the observed linear relation is likely caused in part by inaccuracies in the determination of specific internal surface and IP parameters and discrepancies of scale between the two measurements. However, the scatter also certainly reflects the fact that this simplified relationship fails to account for the full complexity of the IP phenomenon in porous media.

We propose that this linear IP- S_{por} relation can be considered the IP equivalent of the classical Archie empirical relation for resistivity (Archie, 1942). In the same way that Archie's relationship was developed using a large database of resistivity and porosity measurements, we have developed a relationship based on a large database of IP and S_{por} measurements. The Archie relation describes a power-law relation between electrical conductivity and porosity resulting from electrolytic conduction through the available interconnected pore volume whereas our IP- S_{por} relation describes a relation between parameters derived from IP measurements and available pore surface area. The coefficients c_p and c_s describing the slope of a linear relation between the IP parameter (m_n or σ'') and S_{por} depend on the lithology of the investigated formation and the chemistry of the pore-filling fluid. In a similar way, the factor a and the cementation exponent m are used to adopt Archie's power law to a certain lithology.

Given the importance of S_{por} in determining the hydraulic properties of porous media, our findings encourage the wider adoption of IP and SIP as hydrogeophysical exploration methods. The relationships between polarization magnitude and S_{por} presented here could be used to constrain interpretation of field-scale IP data sets aimed at imaging subsurface variations in hydraulic properties.

GLOSSARY OF TERMS

Archie's law: Describes the empirical relation between electrical resistivity ρ and porosity ϕ of a sample by the power law $\rho = a\rho_w\phi^{-m}$, where ρ_w is the fluid resistivity. The cementation exponent m and the factor a depend on lithology.

Global estimate of the polarization magnitude (m_n): Quantifies the effective (or average) polarization magnitude accounting for the

frequency dependence in observed σ'' . Determined from the integration of a sum of individual chargeabilities for a distribution of Debye models spanning the measured frequency range (see Appendix A for details). Represents the polarization magnitude obtained from a multifrequency SIP measurement.

Imaginary conductivity (σ''): The out-of-phase component of the measured complex conductivity (σ^*) associated with charge storage. Computed for a single frequency as $\sigma'' = |\sigma|\sin\varphi$, where $|\sigma|$ and φ are the measured conductivity magnitude and phase shift, respectively, recorded with frequency-domain IP instruments. Represents the polarization magnitude obtained from a single-frequency IP measurement.

Polarizability of the mineral surface per unit S_{por} : A concept proposed here to explain differences in the polarization magnitude (σ'' or m_n) between samples with equal S_{por} . Assumed to represent the lump effects of interfacial chemistry of the surface (rather than size of the surface) in determining the polarization magnitude.

Specific surface area per unit mass (S_m): The amount of mineral surface S per matrix mass m of the investigated sample ($S_m = S/m$) as typically measured with a gas-adsorption technique. Normally reported as m^2/g .

Specific surface area per unit volume (S_{tot}): The surface area S normalized by the total sample volume V of the sample. Calculated from $S_{\text{tot}} = S/V = S_m \cdot d_s \cdot (1 - \phi)$, where d_s is the solid mineral density and ϕ the porosity of the sample.

Specific surface area per unit pore volume (S_{por}): The surface area S normalized by the pore volume V_p of the sample. Calculated from $S_{\text{por}} = S/V_p = S_m \cdot d_s \cdot \frac{1-\phi}{\phi} = S_{\text{tot}}/\phi$.

APPENDIX A

DEBYE DECOMPOSITION

The frequency-dependent impedance $Z(\omega)$ that describes an IP spectrum can be fitted by an overlay of n Debye terms

$$Z(\omega) = R_0 \left(1 - \sum_{k=1}^n m_k \left(1 - \frac{1}{1 + i\omega\tau_k} \right) \right), \quad (\text{A-1})$$

where R_0 is the direct-current (DC) resistance, m_k and τ_k are chargeability and relaxation time of a single Debye model, respectively, and ω the angular frequency. Equation A-1 is equivalent to the time-domain voltage decay $U(t)$ normalized to the constant strength of current I

$$Z(t) = \frac{U(t)}{I} = R_0 \sum_{k=1}^n m_k \exp\left(\frac{-t}{\tau_k}\right). \quad (\text{A-2})$$

The n pairs of chargeability m_k and relaxation time τ_k and the value of the DC resistance R_0 are the resulting parameters of a fitting procedure (Nordsiek and Weller, 2008) considering real and imaginary parts of the measured impedance. To compress the amount of data, we propose four integrating parameters:

- 1) the total chargeability $m_t = \sum_{k=1}^n m_k$
- 2) the mean relaxation time $\bar{\tau} = \exp\left(\frac{\sum_{k=1}^n m_k \ln(\tau_k)}{m_t}\right)$,
- 3) the nonuniformity parameter $U_\tau = \frac{\tau_{60}}{\tau_{10}}$ with τ_{10} and τ_{60} marking those relaxation times whereby in a cumulative curve 10% and 60% of the total chargeability is reached

- 4) the DC resistivity $\rho_0 = R_0 K$ with K being the geometric factor of the used electrode array

The normalized chargeability m_n derived from a DD results from the ratio between total chargeability m_t and DC resistivity ρ_0

$$m_n = \frac{m_t}{\rho_0}. \quad (\text{A-3})$$

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